

EXAM 7 – SPRING 2013

21. (1.25 points)

a. (0.75 point)

Within the context of Basel II, describe how a firm's mix of business affects the basic indicator approach and the standardized approach to calculating the required capital for operational risk.

b. (0.5 point)

Describe a feature of the standardized approach that could make a firm prefer it to the basic indicator approach.

CONTINUED ON NEXT PAGE

Question 21 Sample Answer

Solution 1

- a) The basic indicator applies a single factor to the entire business without giving consideration to the mix of business. Specially, apply a factor of $\alpha = .15$ to the entire entity (3 year average gross income). The standardized approach, on the other hand, applies a different factor, β , to different lines of business, specifically for retail banking, $\beta=.12$, for commercial banking, $\beta=.15$, and for everything else, $\beta=.18$. So the standardized approach does account for the firm's mix of business by applying different factors to different business segments while the Basic Indicator does not.
- b) The standardized approach effectively gives credit for years with negative (or \$0) gross income, while the basic indicator does not. For the basic indicator, multiply α times 3 years gross income average, but years with negative or \$0 gross income are excluded from average (so if one year is negative, will only average over two other years). But for the standardized approach, add up all years (use \$0 for negative years), and divide by three (even if \$0 is used for some years). This can lower the capital requirement if the firm had a year with negative gross income.

Solution 2

- a) Basic Approach = $[1 / Z_n \times \sum \text{from } i=1 \text{ to } i=3 \text{ of } \max(\text{Gross Income}, 0)] \times \alpha$

Where $Z_n = \#$ years where gross income > 0 for the last 3 years. The mix of business doesn't have an effect on the basic approach as the factor α is constant (15%) for all lines.

Standardized approach = $1/3 \times \sum \max(\sum \beta_j GI_j, 0)$. Here the mix of business affects the capital as each line of business has a specific β (usually in the range of 12-18%) that is applied to the Gross Income for that line (e.g., Retail Banking: $\beta = 12\%$, Commercial Banking: $\beta = 15\%$, Other: $\beta = 18\%$). The capital requirement therefore depends on the mix.

- b) The standardized approach allows negative gross income in one area to be offset by positive gross income in another line by year. Also, the standardized approach always averages the last 3 years, but the Basic only divides by $\#$ of years where Gross Income > 0 . So, this may result in a lower capital need using the Standardized Approach.

Examiner Comments

Part a) of the problem required understanding of a key difference in how capital requirements are done under two methods described in the syllabus reading and how mix of business affected the capital calculations. Candidates were allowed to demonstrate that understanding either through formula presentation or narrative descriptions, although often both forms of response were entered in the candidate solutions. In those cases where candidates entered a solution but did not receive full credit, it was generally because the response was vague or incomplete in some respect, the candidate failed to indicate that capital charges varied by type of business in the standardized approach, or else the candidate did not explicitly answer the question as to whether mix of business impacted the capital calculation under each method.

Part b) of the problem could be answered in several ways, any one of which could receive full credit. Common reasons for not assigning full credit to a response in this part included insufficient explanation as to why positive and negative income offsets (or “netting”) across lines of business within a given year could generate different indicated capital amounts under the standardized method, or failure to describe what circumstances were necessary to provide a benefit under standardized approach’s unique capital formula (e.g., an overall loss in one year or a higher capital charge on a line of business with a loss).